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3. Let C be the set of complex numbers. Define addition on C by

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

and define scalar multiplication by

$$\alpha(a + bi) = \alpha a + \alpha bi$$

for all real numbers α . Show that C is a vector space with these operations.

Let $(x, y, z) \in C$, $x = a + bi$, $y = c + di$, $z = e + fi$, $(a, b, c, d, e, f) \in \mathbb{R}$
 we know $(a+c)$, $(b+d)$ and $(e+f) \in \mathbb{R}$ because $\mathbb{R}$ are closed under addition. Thus, $x+y \in C$

$$\begin{aligned} A1. \quad x + y &= (a + bi) + (c + di) = (a + c) + (b + d)i = (c + a) + (d + b)i = (c + di) + (a + bi) \\ &= y + x \\ \Rightarrow x + y &= y + x \end{aligned}$$

$$\begin{aligned} A2. \quad x + (y + z) &= (a + bi) + [(c + di) + (e + fi)] = (a + bi) + [(c + e) + (d + f)i] \\ &= (a + (c + e)) + (b + (d + f))i = (a + c + e) + (b + d + f)i \\ &= ((a + c) + e) + ((b + d) + f)i = ((a + c) + (b + d)i) + (e + fi) \\ &= ((a + bi) + (c + di)) + (e + fi) = (x + y) + z \end{aligned}$$

A3.

There exist an element $0 = 0 + 0i$ s.t. $X + 0 = (a + bi) + (0 + 0i)$
 $= (a + 0) + (b + 0)i = a + bi = X$

A4.

exist $-X = -a + (-b)i$ s.t. $X + (-X) = (a + bi) + (-a + (-b)i)$
 $= (a + (-a)) + (b + (-b))i = 0 + 0i = 0$

A5.

$\alpha \in \mathbb{R}$

$$\begin{aligned}\alpha(X + Y) &= \alpha((a + bi) + (c + di)) = \alpha((a + c) + (b + d)i) = \alpha(a + c) + \alpha(b + d)i \\ &= (\alpha a + \alpha c) + (\alpha b + \alpha d)i = (\alpha a + \alpha b i) + (\alpha c + \alpha d i) \\ &= \alpha(a + bi) + \alpha(c + di) = \alpha X + \alpha Y\end{aligned}$$

A6.

$$\begin{aligned}(\alpha + \beta)X &= (\alpha + \beta)(a + bi) = (\alpha + \beta)a + (\alpha + \beta)b i = (\alpha a + \beta a) + (\alpha b + \beta b)i \\ &= (\alpha a + \alpha b i) + (\beta a + \beta b i) \\ &= \alpha(a + bi) + \beta(a + bi) \\ &= \alpha X + \beta X\end{aligned}$$

A7

$$\begin{aligned}
 (\alpha\beta)x &= (\alpha\beta)(a+bi) = (\alpha\beta a + \alpha\beta bi) = (\alpha(\beta a) + \alpha(\beta bi)) \\
 &= \alpha(\beta a + \beta bi) = \alpha(\beta(a+bi)) \\
 &= \alpha(\beta x)
 \end{aligned}$$

A8. exist $\alpha = 1 \in \mathbb{R}$ st. $\alpha \cdot x = 1 \cdot (a+bi) = (1 \cdot a + 1 \cdot bi) = a+bi = x$

$\mathbb{C}$ satisfies axioms A1. to A8 so $\mathbb{C}$ is a vector space. ~~xi~~

2. Determine whether the following sets form subspaces of $\mathbb{R}^3$:

(a) $\{(x_1, x_2, x_3)^T \mid x_1 + x_3 = 1\}$

(b) $\{(x_1, x_2, x_3)^T \mid x_1 = x_2 = x_3\}$

(c) $\{(x_1, x_2, x_3)^T \mid x_3 = x_1 + x_2\}$

(d) $\{(x_1, x_2, x_3)^T \mid x_3 = x_1 \text{ or } x_3 = x_2\}$

(a) $S = \{(x_1, x_2, x_3)^T \mid x_1 + x_3 = 1\}$

Let $a = (a_1, a_2, a_3) = (0, 0, 1)$, $b = (b_1, b_2, b_3) = (0.5, 0, 0.5)$

$$a_1 + a_3 = b_1 + b_3 = 1, \quad a \text{ and } b \in S$$

$$a + b = (0.5, 0, 1.5), \quad (a_1 + b_1) + (a_3 + b_3) = 2 \neq 1$$

Thus, $a + b \notin S$. S is not a subspace.

(not closed under linear combinations)

(b)

$$S = \{(x_1, x_2, x_3)^T \mid x_1 = x_2 = x_3\}$$

Let $x_1 = x_2 = x_3 = 0$. the zero vector $(0, 0, 0)^T \in S$, so S is non-empty

let $a = (a_1, a_2, a_3)^T \in S$, $b = (b_1, b_2, b_3)^T \in S$, $(\alpha, \beta) \in \mathbb{R}$

$$\alpha a + \beta b = (\alpha a_1 + \beta b_1, \alpha a_2 + \beta b_2, \alpha a_3 + \beta b_3)^T$$

$$\left. \begin{array}{l} a_1 = a_2 = a_3 \\ \Rightarrow \alpha a_1 = \alpha a_2 = \alpha a_3 \\ b_1 = b_2 = b_3 \\ \Rightarrow \beta b_1 = \beta b_2 = \beta b_3 \end{array} \right\} \begin{array}{l} \Rightarrow \alpha a_1 + \beta b_1 = \alpha a_2 + \beta b_2 = \alpha a_3 + \beta b_3 \\ \text{This shows } \alpha a + \beta b \in S, \text{ addition} \\ \text{and scalar multiplication are satisfied.} \end{array}$$

$$(c) \quad S = \left\{ (x_1, x_2, x_3)^T \mid x_3 = x_1 + x_2 \right\}$$

Let $x_1 = x_2 = 0$, $x_3 = 0 + 0 = 0$. The zero vector $(0, 0, 0)^T \in S$, so S is non-empty.

Let $a = (a_1, a_2, a_3)^T$, $b = (b_1, b_2, b_3)^T$. $a, b \in S$, $\alpha, \beta \in \mathbb{R}$

$$a_3 = a_1 + a_2, \quad b_3 = b_1 + b_2$$

$$\alpha a = (\alpha a_1, \alpha a_2, \alpha a_3)^T, \quad \beta b = (\beta b_1, \beta b_2, \beta b_3)^T$$

$$\alpha a + \beta b = (\alpha a_1 + \beta b_1, \alpha a_2 + \beta b_2, \alpha a_3 + \beta b_3)^T$$

$$\alpha a_3 + \beta b_3 = \alpha(a_1 + a_2) + \beta(b_1 + b_2) = \alpha a_1 + \alpha a_2 + \beta b_1 + \beta b_2$$

$$\Rightarrow (\alpha a_3 + \beta b_3) = (\alpha a_1 + \beta b_1) + (\alpha a_2 + \beta b_2)$$

(addition + scalar multiplication)

This shows S is closed under linear combination,

S is a subspace $\#$

$$(d) \quad S = \left\{ x_1, x_2, x_3 \mid x_3 = x_2 \text{ or } x_3 = x_1 \right\}$$

Let $a = (a_1, a_2, a_3) = (0, 1, 1)$. $b = (b_1, b_2, b_3) = (1, 0, 1)$

$$\alpha, \beta \in \mathbb{R}, \quad \alpha = 2, \quad \beta = 1$$

$$\alpha a + \beta b = (0, 2, 2) + (1, 0, 1) = (1, 2, 3)$$

$1 \neq 3$ and $1 \neq 2 \Rightarrow S$ is not closed under linear combinations

Thus, S is not a subspace $\#$

13. Given

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix},$$

$$\mathbf{x} = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix}$$

(a) Is $\mathbf{x} \in \text{Span}(\mathbf{x}_1, \mathbf{x}_2)$?

(b) Is $\mathbf{y} \in \text{Span}(\mathbf{x}_1, \mathbf{x}_2)$?

Prove your answers.

We know row operations wouldn't change the linear combination of columns in a matrix, so we can check whether vector is in one subspace by Gaussian elimination.

(a)

$$X = \left[\begin{array}{cc|c} 1 & 1 & -1 \\ 2 & 3 & 2 \\ -1 & -2 & -1 \end{array} \right] = \left[\begin{array}{cc|c} 1 & 1 & -1 \\ 2 & 3 & 2 \\ -1 & -2 & -1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cc|c} 1 & 1 & -1 \\ 0 & 1 & 4 \\ 0 & -1 & -2 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & -1 \\ 0 & 1 & 4 \\ 0 & 0 & 2 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{array} \right], \text{ no solution}$$

As we can see, $\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$ cannot be shown by linear combinations of $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ (column₁) and $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ (column₂).

Thus, $\mathbf{x} \notin \text{Span}(\mathbf{x}_1, \mathbf{x}_2)$

(b)

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ x_1 & x_2 & y \\ 1 & 1 & 1 \end{array} \right] = \left[\begin{array}{cc|c} 1 & 1 & -1 \\ 2 & 3 & 2 \\ -1 & -2 & -3 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & -1 \\ 0 & 1 & 4 \\ 0 & -1 & -4 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cc|c} 1 & 1 & -1 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & -5 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{array} \right]$$

By this result, we know $-5 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -5 \\ 4 \\ 0 \end{bmatrix}$, which means

$$-5x_1 + 4x_2 = y \Rightarrow \begin{bmatrix} -5 \\ -10 \\ 5 \end{bmatrix} + \begin{bmatrix} 4 \\ 12 \\ -8 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} = y$$

Thus $y \in \text{Span}(x_1, x_2)$ ~~✗~~

7. Let $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ be linearly independent vectors in $\mathbb{R}^n$ and let

$$\mathbf{y}_1 = \mathbf{x}_2 - \mathbf{x}_1, \quad \mathbf{y}_2 = \mathbf{x}_3 - \mathbf{x}_2, \quad \mathbf{y}_3 = \mathbf{x}_3 - \mathbf{x}_1$$

Are $\mathbf{y}_1$, $\mathbf{y}_2$, and $\mathbf{y}_3$ linearly independent? Prove your answer.

If y_1, y_2, y_3 are linearly independent, they satisfy

$$c_1 y_1 + c_2 y_2 + c_3 y_3 = 0 \text{ and } c_1 = 0, c_2 = 0, c_3 = 0$$

$$\Rightarrow \begin{bmatrix} | & | & | \\ y_1 & y_2 & y_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} | \\ c \\ | \end{bmatrix} = 0, \text{ which also means nullspace of } \begin{bmatrix} | & | & | \\ y_1 & y_2 & y_3 \\ | & | & | \end{bmatrix} \text{ is } 0 \text{ (trivial solution)}$$

$$c_1(x_2 - x_1) + c_2(x_3 - x_2) + c_3(x_3 - x_1) = 0$$

$$\Rightarrow (-c_1 - c_3) \cdot x_1 + (c_1 - c_2) x_2 + (c_2 + c_3) x_3 = 0$$

Since x_1, x_2, x_3 are linearly independent, $-c_1 - c_3 = c_1 - c_2 = c_2 + c_3 = 0$

$$\begin{bmatrix} -1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} -1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 0 & -1 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

We notice there is a free variable in the solution, which means this system has infinite solutions, and y_1, y_2, y_3 are not linearly independent. (one of the solutions like $(1, 1, -1)$)

20. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be linearly independent vectors in a vector space V . Show that $\mathbf{v}_2, \dots, \mathbf{v}_n$ cannot span V .

Assume $\mathbf{v}_2, \dots, \mathbf{v}_n$ can span V , and $\mathbf{v}_1 \in V$, so we can find

$$\mathbf{v}_1 = c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 + \dots + c_n \mathbf{v}_n$$

$$\Rightarrow \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 + \dots + c_n \mathbf{v}_n = \mathbf{0}$$

(nullspace must be not 0)

By this equation, we can find the coefficient of $\mathbf{v}_1$ is always 1, which means $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n$ are linearly dependent. This contradicts the assumption that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent.

Thus, $\mathbf{v}_2, \dots, \mathbf{v}_n$ cannot span V . ~~✗~~

10. The vectors

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix},$$

$$\mathbf{x}_3 = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{x}_4 = \begin{bmatrix} 2 \\ 0 \\ 6 \end{bmatrix}, \quad \mathbf{x}_5 = \begin{bmatrix} 0 \\ 4 \\ 5 \end{bmatrix}$$

span $\mathbb{R}^3$. Pare down the set $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5\}$ to form a basis for $\mathbb{R}^3$.

$$X = \begin{bmatrix} | & | & | & | & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \mathbf{x}_3 & \mathbf{x}_4 & \mathbf{x}_5 \\ | & | & | & | & | \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 0 & 2 & 2 & 0 & 4 \\ 3 & 1 & -1 & 6 & 5 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 0 & 2 & 2 & 0 & 4 \\ 0 & -5 & -5 & 0 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 0 & 2 & 2 & 0 & 4 \\ 0 & 0 & 0 & 0 & 15 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -1 & 2 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

pivots are located in x_1, x_2, x_5 and x_1, x_2, x_5 are

linearly independent, $\text{rank}(X) = \dim(X) = 3$. Hence, (x_1, x_2, x_5)

are enough to span $\mathbb{R}^3$. Additionally, x_3, x_4 can be shown by linear combination, so they are redundant, $x_3 = -x_1 + x_2$, $x_4 = 2x_1$,

Again, $c_1 x_1 + c_2 x_2 + c_3 x_3 = 0$ from previous elimination

$$\begin{bmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = 0, \quad \begin{matrix} c_1 = 0 \\ c_2 = 0 \\ c_3 = 0 \end{matrix}$$

x_1, x_2, x_3 are linearly independent

So the basis for $\mathbb{R}^3$ is $(x_1, x_2, x_3) = \left(\begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 4 \\ 5 \end{bmatrix} \right)$ ✗